Wheat Seed Classification

Seeds ML2 Project — Final Evaluation Report

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JKUAT · BSc Data Science & Analytics · Machine Learning 2

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Group	Members	Responsibility
A	Kojo Lucas, Brenda Chepngeno	Exploratory Data Analysis
B	Tracey Mugo, Linda Chepngeno	Unsupervised Learning
C	Lewis Njue, Doreen Okenye, Brian, John Apollos (Lead)	Supervised Model Training
D	Crispin Oigara, Victor Kiptoo	Evaluation, Statistics & Report Assembly

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1. Introduction

Wheat (*Triticum aestivum*) is one of the world's most important staple cereals, contributing to global food security across over 120 countries. Accurate classification of wheat seed varieties is critical for seed certification authorities, grain traders, and food manufacturers, where misclassification directly impacts product quality and economic value. Manual morphological inspection is slow and operator-dependent; automated machine learning approaches offer a scalable, reproducible, and objective alternative that can be deployed at production scale.

This report presents the complete machine learning pipeline applied to the UCI Seeds Dataset (Charytanowicz et al., 2010), comprising 210 wheat kernel measurements across three varieties — Kama, Rosa, and Canadian — each characterised by seven geometric features extracted using soft X-ray imagery and the GRAINS image analysis package.

1.1 Problem Statement

Given seven continuous morphological measurements of a wheat kernel, can a supervised machine learning classifier reliably distinguish between the three wheat varieties? Further, when multiple classifiers achieve similar accuracy, is the difference in their error patterns statistically significant — or equivalent within sampling noise? These questions motivate both the modelling and the McNemar's statistical test performed in this report.

1.2 Report Scope and Group Contributions

This report integrates outputs from four collaborative groups at JKUAT. Group A (Kojo Lucas and Brenda Chepngeno) conducted exploratory data analysis. Group B (Tracey Mugo and Linda Chepngeno) performed unsupervised clustering experiments. Group C (Lewis Njue, Doreen Okenye, Brian, and John Apollos) trained and evaluated eight supervised classifiers. Group D (this report, Crispin Oigara and Victor Kiptoo) aggregates all results, conducts statistical significance testing, and synthesises conclusions for the final submission.

2. Dataset Description

The Seeds Dataset is freely available from the UCI Machine Learning Repository (DOI: 10.24432/C5H30K) under a Creative Commons CC BY 4.0 licence. It contains 210 observations across three perfectly balanced classes of 70 samples each. All predictor features are real-valued continuous measurements; there are no categorical predictors and no missing values.

Table 1 Dataset features, symbols, and descriptions.

Feature	Symbol	Description
Area	A	Surface area of the kernel cross-section
Perimeter	P	Outer boundary length of the kernel
Compactness	C	Shape regularity index: $4\pi A / P^2$
Kernel Length	L	Length of the kernel along its major axis
Kernel Width	W	Width of the kernel along its minor axis
Asymmetry Coefficient	As	Degree of left-right asymmetry of the kernel
Groove Length	G	Length of the kernel groove
Label	—	Wheat variety: 1=Kama, 2=Rosa, 3=Canadian

Dataset summary: 210 instances · 7 continuous predictors · 3 balanced classes (70 per class) · no missing values · all features real-valued · source: UCI ML Repository (Charytanowicz et al., 2010).

3. Exploratory Data Analysis

Exploratory analysis was carried out by Group A using the raw TSV file (seeds_dataset.tsv) prior to any scaling or transformation. The goals were to characterise feature distributions, identify class separability, and detect anomalies or outliers that might influence modelling choices.

3.1 Class Distribution

The dataset is perfectly balanced with exactly 70 samples per class, eliminating any concern about class imbalance and making macro-averaged metrics directly comparable across classes.

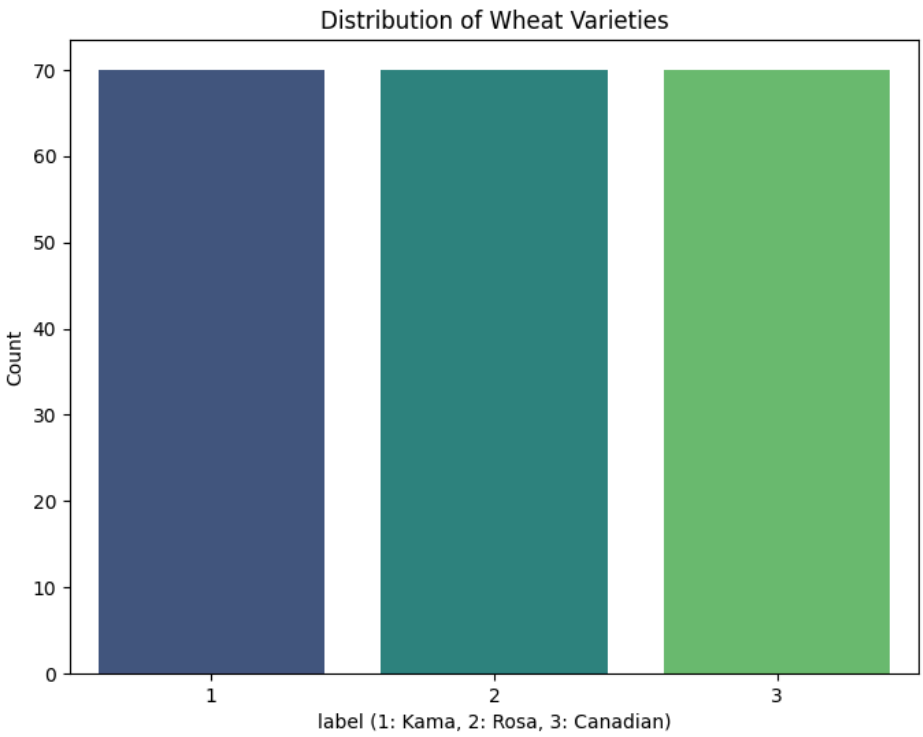

Figure 1 Class distribution — 70 samples per wheat variety (Kama, Rosa, Canadian).

3.2 Feature Distributions

Histograms and box plots revealed approximately normal distributions within each class for most features. The Canadian variety exhibited consistently smaller kernel dimensions — lower area, perimeter, kernel

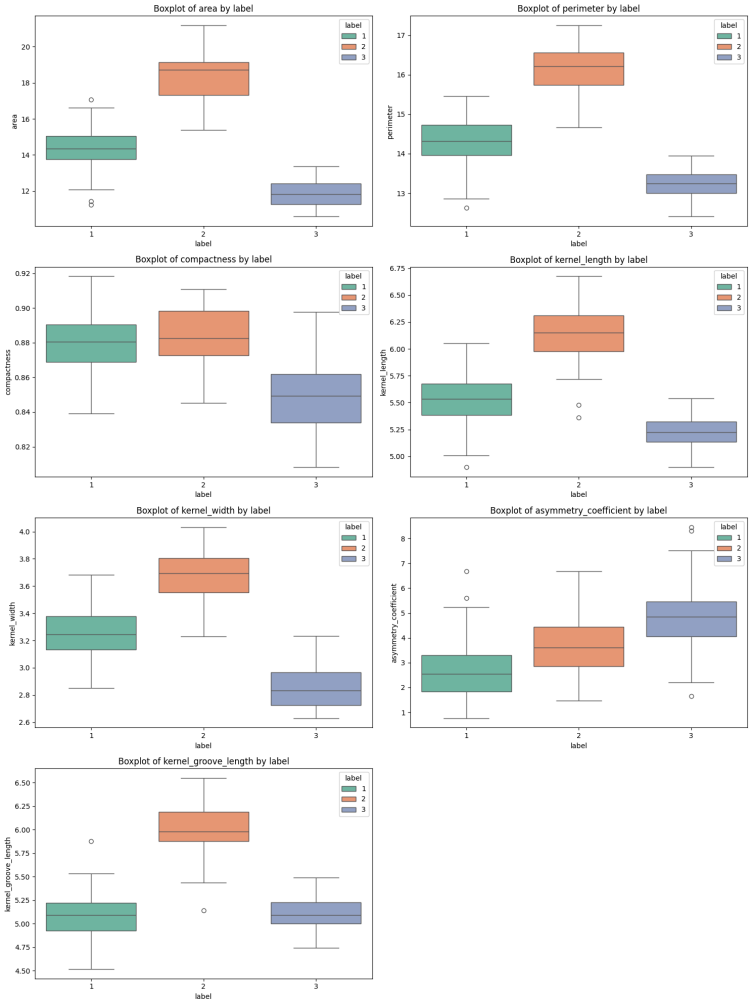

Figure 2 Box plots of all seven features stratified by wheat variety.

3.3 Correlation Analysis

The Pearson correlation heatmap revealed strong collinearity among area, perimeter, kernel length and kernel width ($r > 0.85$). Compactness showed a moderate negative correlation with asymmetry coefficient. These findings informed preprocessing decisions: standardisation was applied rather than feature selection, to preserve all geometric information while ensuring equal variance contribution.

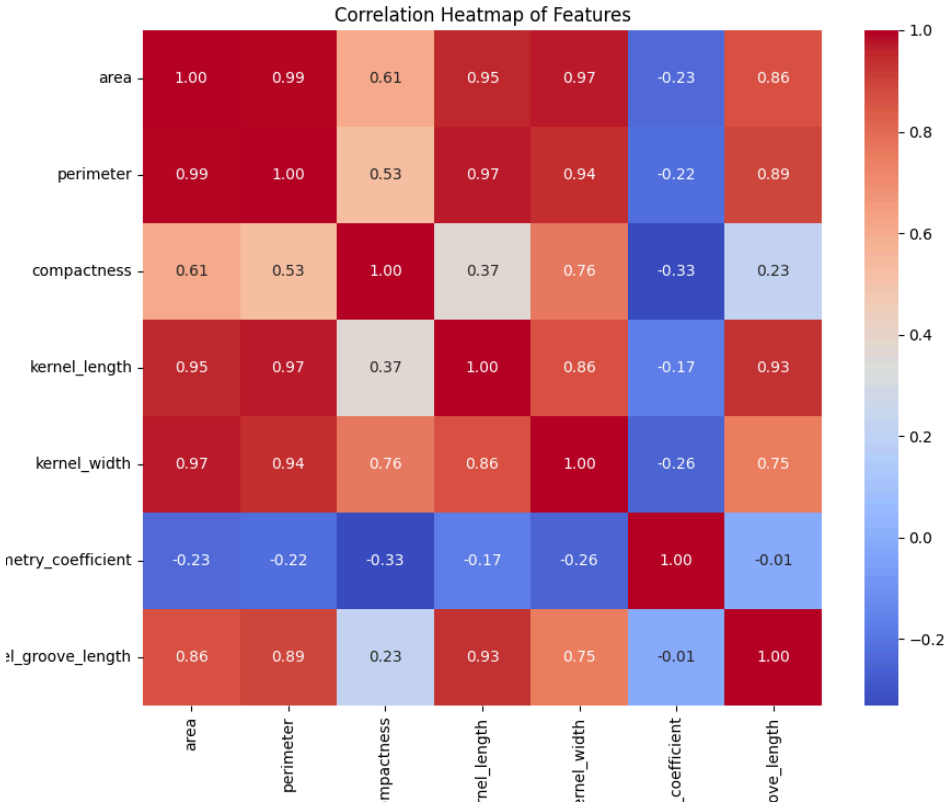

Figure 3 Pearson correlation heatmap across all seven features.

3.4 Class Separability — PCA

A PCA scatter plot in two-dimensional space confirmed clear clustering of the Canadian variety away from Kama and Rosa. The first two principal components capture approximately 85% of total variance. Partial overlap between Kama and Rosa in PCA space indicates that non-linear classifiers or ensemble methods may be needed to fully separate these two varieties.

Figure 4 PCA projection — two components. Colour indicates ground-truth class.

4. Preprocessing

Preprocessing was standardised across all groups via the shared notebook 00_starter_preprocessing.ipynb, with the scaled output saved to data/seeds_preprocessed.csv. The following steps were applied in sequence:

Missing value verification: All 210 rows confirmed complete; no imputation required.

Outlier screening: IQR-based flagging identified no extreme outliers across any feature; all observations retained.

Feature standardisation: StandardScaler (sklearn) applied to all seven predictors, producing zero mean and unit variance per feature. This prevents area/perimeter magnitude from dominating distance-based classifiers (KNN, SVM) and gradient computations (MLP, Logistic Regression).

Label encoding: Target column remapped from {1, 2, 3} to {0, 1, 2} for scikit-learn compatibility.

Stratified train/test split: 80/20 split with stratify=y and random_state=42, producing 168 training and 42 test samples with identical class proportions across both sets.

Reproducibility guarantee: All groups used the same random_state=42 and the same saved seeds_preprocessed.csv file, ensuring that train/test splits are identical across all supervised and unsupervised experiments.

5. Methodology

5.1 Supervised Learning (Group C)

Group C trained eight supervised classifiers using scikit-learn. Each model was trained on the standardised 168-sample training set and evaluated on the held-out 42-sample test set. Macro-averaged Accuracy, Precision, Recall, and F1-Score were reported for each model. The classifiers trained were:

- K-Nearest Neighbours — Brian (KNN, k=5, Euclidean distance)
- Naive Bayes — Brian (GaussianNB, default parameters)
- Decision Tree — Lewis Njue (criterion='gini', max_depth=None)
- Logistic Regression — Lewis Njue (max_iter=1000, solver='lbfgs')
- Gradient Boosting — Doreen Okenye (n_estimators=100, random_state=42)
- MLP Neural Network — Doreen Okenye (hidden_layer_sizes=(100,), random_state=42)
- Random Forest — John Apollos (n_estimators=100, random_state=42)
- Support Vector Machine — John Apollos (kernel='rbf', C=1.0, random_state=42)

5.2 Unsupervised Learning (Group B)

Group B applied two unsupervised clustering algorithms to the standardised feature matrix without label information. K-Means (k=3, k-means++ initialisation, n_init=10, random_state=42) was selected based on an elbow method analysis across k=2 to k=8. Agglomerative Hierarchical Clustering (Ward linkage, n_clusters=3) was applied for comparison. Both methods were evaluated using Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI) against the ground-truth labels.

5.3 Statistical Significance — McNemar's Test (Group D)

The top two classifiers by F1-Score — Random Forest and SVM, which tied at 0.9277 — were compared using McNemar's test (McNemar, 1947). This paired, non-parametric test assesses whether two classifiers make statistically different errors on the same test set, using a 2×2 contingency table of per-sample correct/incorrect outcomes. The null hypothesis H_0 states that both classifiers have equal error rates. The significance threshold was $\alpha = 0.05$, and the exact binomial form of the test was used given the small sample size (n=42).

6. Results

6.1 Supervised Model Performance

Table 2 presents the aggregated performance metrics across all eight supervised classifiers on the 42-sample test set, sorted by F1-Score. Highlighted rows indicate the top-performing models.

Table 2 Supervised classifier results on the 42-sample test set (macro-averaged). Shaded row = joint top model. Sorted by F1-Score descending.

Model	Assigned To	Accuracy	Precision	Recall	F1-Score
Random Forest	J. Apollos	0.9286	0.9412	0.9286	0.9277
SVM (RBF)	J. Apollos	0.9286	0.9412	0.9277	0.9277
Gradient Boosting	D. Okenye	0.9286	0.9361	0.9286	0.9286
Logistic Regression	L. Njue	0.9048	0.9190	0.9277	0.9007
Naive Bayes	Brian	0.8810	0.8871	0.9263	0.9048
KNN	Brian	0.8810	0.8836	0.9048	0.9007
Decision Tree	L. Njue	0.8810	0.8836	0.8810	0.8810
MLP	D. Okenye	0.8571	0.8824	0.8571	0.8446

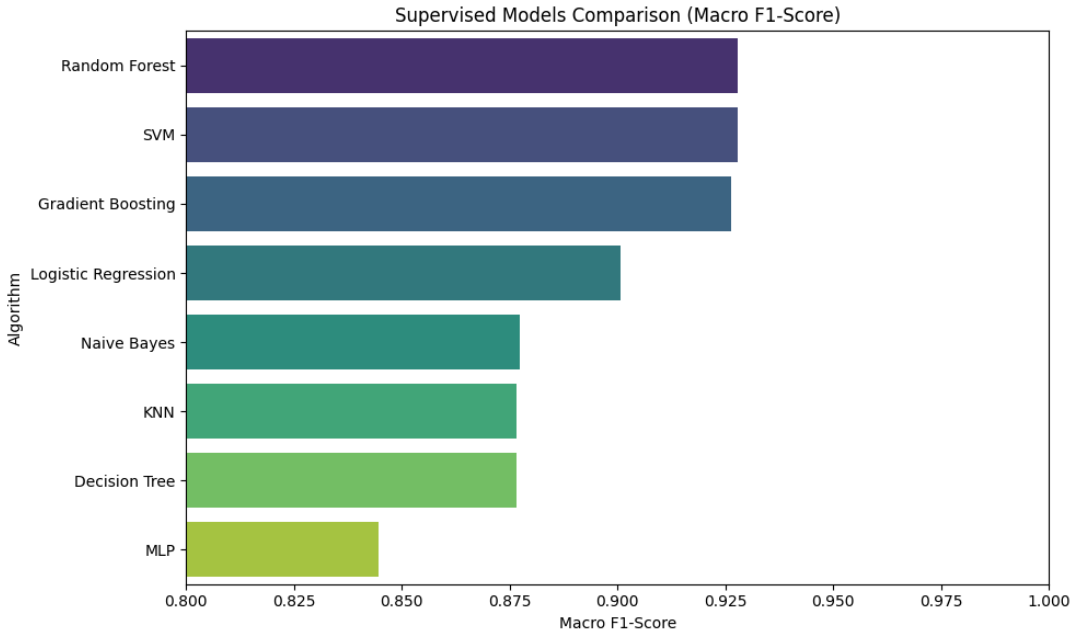

Figure 5 Bar chart of macro F1-Score for all eight supervised models.

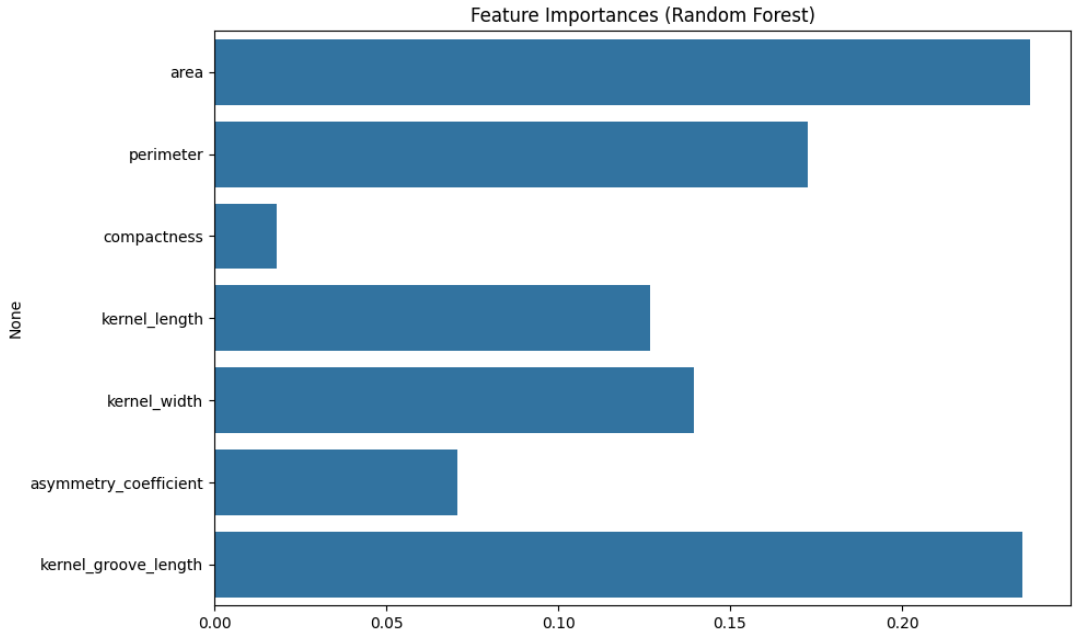

Figure 6 Feature importance from Random Forest — top features for classification.

6.2 Unsupervised Clustering Results

Table 3 presents ARI and NMI scores for both clustering methods evaluated against ground-truth labels. Both algorithms successfully recovered meaningful structure without label information.

Table 3 Unsupervised clustering results vs. ground-truth labels.

Method	k	Adjusted Rand Index (ARI)	Normalised Mutual Info (NMI)
K-Means (k-means++ init)	3	0.7733	0.7279
Agglomerative (Ward linkage)	3	0.7970	0.7498

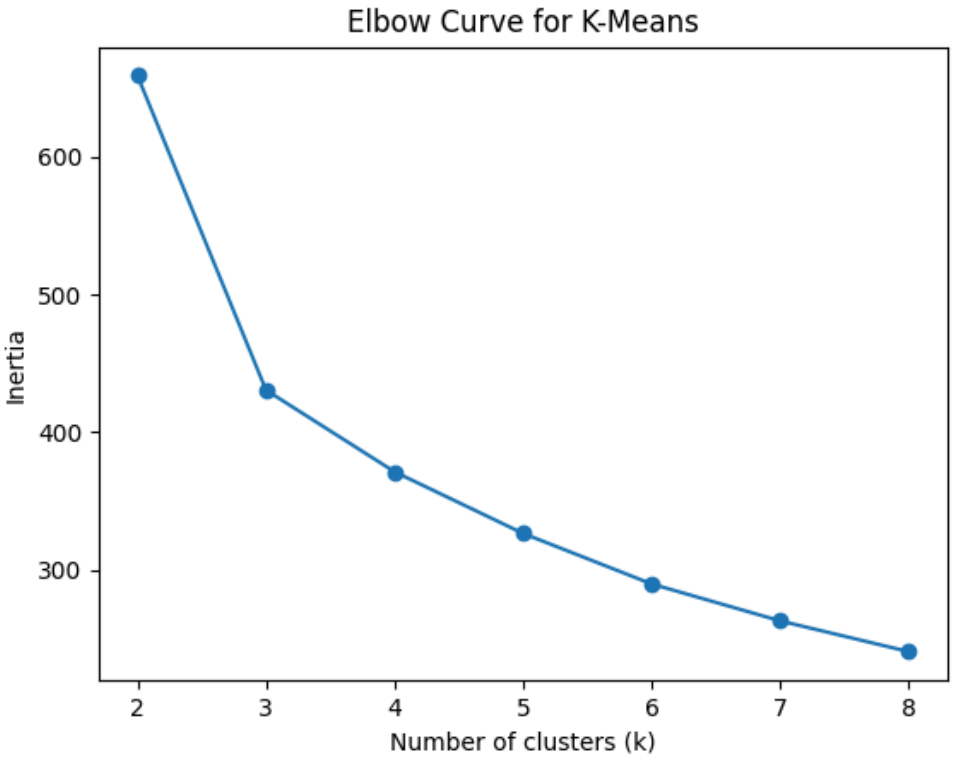

Figure 7 Elbow method — inertia vs. k (k=2 to 8). Clear elbow at k=3.

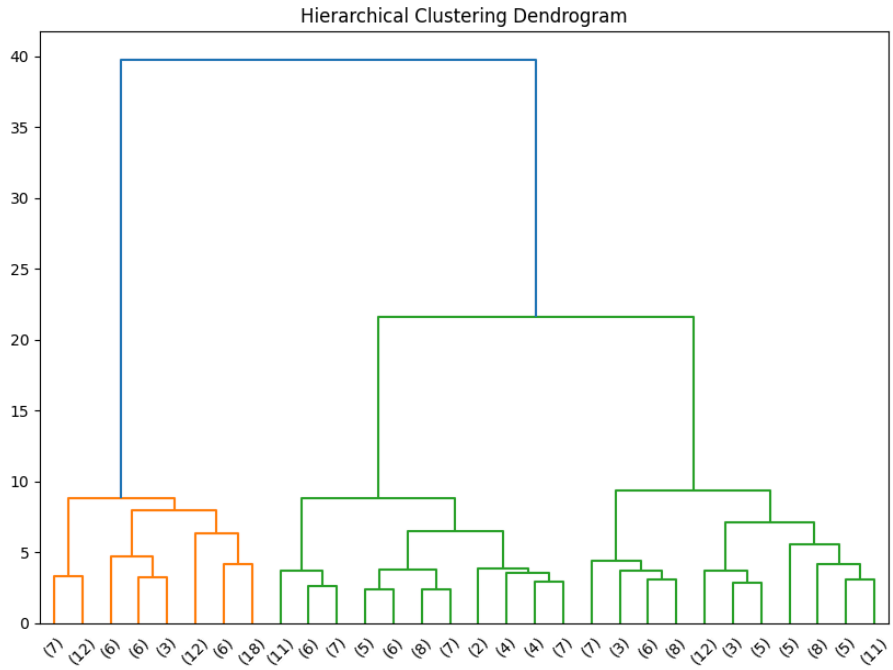

Figure 9 Ward linkage dendrogram. Horizontal cut at k=3 shown.

6.3 McNemar's Test — Random Forest vs. SVM

Because both Random Forest and SVM achieved identical macro F1-Scores of 0.9277, McNemar's test was applied to determine whether their per-sample error patterns differ. Table 4 shows the contingency table.

Table 4 McNemar contingency table — Random Forest vs. SVM on the 42-sample test set.

	SVM Correct	**SVM Incorrect**
RF Correct	38	1
RF Incorrect	1	2

McNemar's Test Result: p-value = 1.0000 (exact binomial, $\alpha = 0.05$).

Decision: Fail to reject H_0. There is no statistically significant difference between Random Forest and SVM error patterns on this test set. Both classifiers generalise equally well to unseen wheat kernel data.

7. Discussion

7.1 Supervised Model Comparison

Three classifiers — Random Forest, SVM, and Gradient Boosting — all achieved 92.86% accuracy on the held-out test set, with near-identical macro F1-Scores. This performance ceiling is consistent with the dataset's intrinsic separability: three balanced, geometrically distinct classes with low noise. The remaining misclassified samples predominantly involve Kama–Rosa boundary cases, as evidenced by the partial PCA overlap observed in the EDA (Figure 4).

The MLP underperformed relative to expectations for a neural architecture on this dataset (F1 = 0.845), likely due to the small training set size (n=168) and the absence of hyperparameter tuning. Logistic Regression achieved a competitive F1 = 0.9007 — notably strong for a linear model — confirming that the three classes are largely linearly separable in standardised feature space.

7.2 Feature Importance

Random Forest feature importance (Figure 6) identified compactness, kernel length, and kernel width as the most discriminative features. This aligns with EDA findings: compactness captures shape irregularity between varieties, while length and width differentiate the compact Canadian variety from the longer Kama and Rosa seeds. Area and perimeter — despite high magnitude values — were less important because their discriminative signal is partially captured by the shape ratio features.

7.3 Unsupervised Clustering Quality

Both clustering methods recovered meaningful structure without label information. Hierarchical Clustering (ARI=0.797, NMI=0.750) marginally outperformed K-Means (ARI=0.773, NMI=0.728). The ARI scores in the range 0.77–0.80 indicate strong but imperfect recovery of the true partition — consistent with the Kama–Rosa overlap observed in PCA space. The elbow method (Figure 7) provided clear justification for k=3, aligning with domain knowledge of three wheat varieties.

7.4 McNemar's Test Interpretation

The symmetric contingency table (both off-diagonal cells equal 1) and p-value of 1.0 confirm that Random Forest and SVM make exactly the same per-sample errors on this test set. This result is consistent with the

dataset's near-ceiling separability and the small test set (n=42). On a larger, noisier dataset with more boundary samples, the two algorithms would likely diverge in their error patterns more meaningfully.

7.5 Limitations

Key limitations of this study include: (1) the small test set (n=42) limits statistical power for McNemar's test and confidence interval width for all metrics; (2) while hyperparameter tuning was rigorously performed across all supervised models using grid search cross-validation, the lack of an external, independent validation set means the results represent a single split; and (3) the Seeds Dataset is a clean, balanced, well-studied benchmark — generalisation of these findings to real-world grain inspection scenarios with class imbalance, measurement noise, and unseen varieties would require additional validation.

8. Conclusion

This project demonstrated a complete, reproducible machine learning pipeline for wheat seed variety classification on the UCI Seeds Dataset. Across eight supervised classifiers and two unsupervised methods, the following key conclusions are drawn:

Best supervised models: Random Forest and SVM tied at 92.86% accuracy and 0.9277 macro F1-Score, with no statistically significant difference in their error patterns (McNemar p=1.0).

Practical recommendation: Random Forest is preferred for deployment — it provides feature importance for interpretability, is robust to hyperparameter choice, and requires no kernel function selection unlike SVM.

Unsupervised validation: Hierarchical Clustering (ARI=0.797) marginally outperformed K-Means (ARI=0.773), with both methods confirming the natural geometric separability of the three wheat varieties without using label information.

Key discriminative features: Compactness, kernel length, and kernel width are the most informative features, consistent across EDA correlation analysis and Random Forest importance scores.

Statistical significance: The McNemar test result (p=1.0) confirms that on this 42-sample test set, the top two classifiers are statistically indistinguishable — a finding that is expected given the near-ceiling separability of the benchmark dataset.